

Lesson 3: Is a force real?

The Brake · Newton's first and second laws · 6 periods of 50 minutes

Grade 11 scientific track (adaptable to Grade 10 and to equivalent grades in other Arab countries). Physics in English or French; discussion may be in Arabic. The student lesson is free at sababphysics.com: Lessons, Grade 11, Lesson 3.

Learning objectives

- Tell apart describing a motion (kinematics) and explaining it with its causes (dynamics). (A1)
- Determine a mass as the constant ratio force ÷ acceleration from experimental data. (A2)
- State Newton's second law in vector form, $\Sigma F = m \cdot a$, with a in the direction of the net force. (A3)
- Represent the forces on a body and project the law on the direction of motion to calculate an acceleration. (A4)
- Explain the braking passenger using only real forces, exerted by an identified object. (C1)
- Argue whether a force is real or a description, answering the strongest opposite view. (C2, C3)

Competencies of the 2026 framework

Competency	Where it is developed
Critical thinking	Steps 2 and 6: position, argument, position shift
Inquiry	Steps 3 and 4: collect data, find the law
Digital	Virtual lab and data table
Communication and collaboration	Community of inquiry, written reflection
Creativity and innovation	Step 5: Arduino lab or egg-passenger challenge

The six steps at a glance

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| 1 Hook | Film "The Brake" and the taxi simulation. Students predict what pushes the passenger forward. |
| 2 Question | "Is a force real?" Four thinkers, four answers. Students take a position. |
| 3 Investigate | Virtual cart: vary the hanging mass and the cart mass, collect accelerations, compute $F \div a$. |
| 4 Model | Test four models against the data; Newton's second law emerges. Back to the taxi. |
| 5 Make | Arduino lab (measure a real acceleration) or the egg-passenger challenge. |
| 6 Reflect | Position shift, concept ladder, written reflection. |

Before you start: prerequisites

From Chapter 7 and earlier grades. Check them in period 1; if many students are unsure, add a 15-minute review.

- Vectors: add two vectors, and project a vector on two perpendicular axes.
- Velocity and acceleration vectors; $a = \Delta v / \Delta t$.
- Uniformly varied rectilinear motion: $v = a \cdot t + v_0$ and $x = \frac{1}{2} \cdot a \cdot t^2 + v_0 \cdot t + x_0$.
- Forces from earlier grades: weight $P = m \cdot g$, normal reaction, tension, friction, and how to draw them.

Timeline: six 50-minute periods

P1 · 0'	Hook: play the film and the taxi animation on the projector. Students predict individually, then vote.
P1 · 15'	Describe or explain? Students sort four sentences; name kinematics and dynamics only afterwards.
P1 · 35'	Prerequisite check: project a force on two axes; read a velocity and an acceleration from a table.
P2 · 0'	Community of inquiry (circle): "Is a force real?" Read the four thinkers. Roles: facilitator, questioner, summariser, skeptic.
P2 · 35'	Students write their position and one reason, then answer the strongest argument from the other side (step 2). Homework: 3 lab runs on a phone.
P3 · 0'	Investigate (step 3): groups run the virtual lab, vary the force and the mass, and fill the table, including the $F \div a$ column.
P3 · 35'	Groups compare their data: for one cart, $F \div a$ stays constant and equals its mass.
P4 · 0'	Model (step 4): test the four candidate models against the data, then name the law $\Sigma F = m \cdot a$.
P4 · 25'	Explain the taxi with real forces only. The crate: represent the four forces and project on the direction of motion ($F - f = m \cdot a$).
P5 · 0'	Make (step 5): half the class runs the Arduino lab, half builds the egg-passenger cart.
P5 · 40'	Groups swap findings: when the measurement and the theory disagree, why?
P6 · 0'	Exam-style practice: the quiz of the lesson, corrected together.
P6 · 30'	Reflection and position shift (step 6). Students send their reflection to the class group.

The four thinkers (step 2)

Aristotle

4th century BCE

A moving body needs a mover. Without a continuing push, motion stops. For a thrown stone, the air keeps pushing it (*Physics*, VII and VIII).

Ibn Sina (Avicenna)

11th century

A thrown body receives an inclination (mayl) from the thrower that keeps it moving until resistance wears it away. In a void, it would never stop (*The Book of Healing*).

Isaac Newton

1687

A body keeps its state of rest or of straight, uniform motion unless a net force acts. Force changes motion: net force = $m \cdot a$.

Ernst Mach

19th century

Mass and force are defined by the accelerations bodies give each other. Force is a name for mass $\times$ acceleration, not a hidden cause (*The Science of Mechanics*, 1883).

Short summaries for discussion, not quotations.

Common misconceptions and how to challenge them

Motion needs a force.

Ask: what keeps a puck sliding on ice? Show the virtual cart with no friction.

A moving object "has" a force that runs out.

Compare with Ibn Sina's mayl, then ask what the data from step 3 shows.

Heavier objects always accelerate less, whatever the force.

Point to runs where the force changes with the mass.

The forward "push" felt when braking is a real force.

Ask: what object exerts it? No object, no force: the passenger keeps moving while the taxi slows.

Make step: Arduino lab

- Per group: Arduino Uno, HC-SR04 distance sensor, 4 jumper wires, toy car, string, small water bottle.
- Students paste the Serial Monitor data; the platform fits $x(t)$ and compares a with the theory.
- No kits? Use the Wokwi online simulator or the built-in sample data.
- Discussion: when theory and measurement disagree, which one is "wrong"?

Make step: egg-passenger challenge

- Recycled materials only: bottles, cardboard, sponges, tape.
- Teams predict with the on-screen calculator, build, test, then improve once.
- Score: egg intact (3), prediction matched the test (2), design journal uses $F = m \cdot a$ (3), poster (2).
- Citizenship link: display the posters in real service taxis.
- Safety: raw eggs, low ramps, clean up spills at once.

Facilitation moves

- "What do you mean by...?" (clarify)
- "How do you know?" (evidence)
- "Can anyone give a counter-example?"
- "What would Ibn Sina say to Aristotle?"
- Aristotle said the air keeps pushing a thrown stone. Ask: what would happen in a vacuum?
- Do not give the answer before step 4.

Assessment rubric: written argument (steps 2 and 6)

Criterion	1	2	3
Claim	Unclear	Clear	Clear and nuanced
Evidence	None	Daily-life example	Uses own lab data
Reasoning	Restates	Links evidence to claim	Uses Newton's laws correctly
Counter-argument	Absent	Mentioned	Answered

Student worksheet · Lesson 3

Photocopy for each student

Name: _____ Class: _____ Date: _____

1. Predict: what pushes the passenger forward when the taxi brakes?

- ☐ A The braking pushes the passenger forward.
- ☐ B Nothing pushes them. They keep moving while the car slows down.
- ☐ C The car's motion stays in the passenger and carries him forward.
- ☐ D The air inside the car pushes them.

2. My data (step 3)

Run	Hanging mass (g)	Cart mass (g)	Net force F (N)	Acceleration a (m/s ²)	F ÷ a (kg)
1					
2					
3					
4					
5					

3. My position: is a force a real thing, or only a description?

A real thing

Only a description



Mark B (before, step 2) and A (after, step 6) on the line.

4. Exit ticket

A passenger lurches forward when a service taxi brakes suddenly. Explain why, using only real forces (forces exerted by an object).
